

Probability & Statistics 5.1) Representation of Data.

Types of Data



Quantitative
numerical data
which can be
further classified
into:

Qualitative
non-numerical data
representing categories
or groups (colours, names)

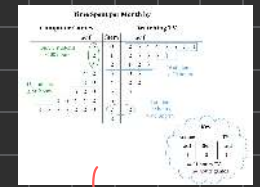
- Discrete data: takes only certain specific values (number of students, dice roll outcome)
- Continuous data: can take any value within a given range (eg. height, weight)

Methods of Data Representation

- #1: Stem and Leaf Diagrams: displays data in ordered rows with equal intervals. Can be used to represent both data points & groups.
- #2: Back-to-Back Stem leaf diagram: A variation of Stem-and-leaf diagram where two data sets are compared side-by-side.
- #3: Histograms: Graphically represents frequency distributions. Column Areas are proportional to frequency, Y-axis is frequency density.

0	9
1	2 5 7
2	2 2 2 3 4 7 8 8 8
3	5 7 8
4	1

Key: 3|1 means 31



back to back stem leaf

Measures of Central tendency

- #1: Mode: Value that occurs most frequently in a data set. Modal class for grouped data: Class Interval with highest frequency.
- #2: Median: for ungrouped data, it is the middle value when data is in ascending data. for grouped data its estimated from cumulative frequency graphs.
- #3: Mean ($\bar{x}$): for ungrouped data, the sum of all data values divided by number of data points for grouped data, Sum of (class midpoint $\times$ frequencies) divided by total number of values.

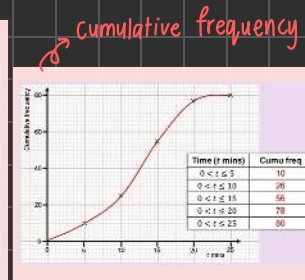
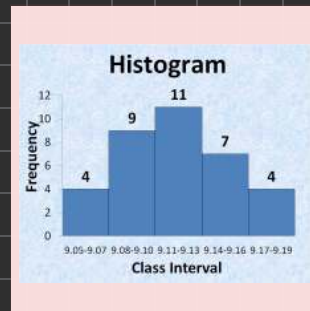
Coded data: Adjusting values of data by using transformations like $+$, $-$, $\times$, $\div$ by a factor.

$$x_{\text{coded}} = \frac{x - b}{a} \quad \left\{ \begin{array}{l} \text{grouped} \\ x: \text{Original values} \\ \bar{x}: \text{mean of original} \\ x_{\text{coded}}: \text{Coded Values} \end{array} \right.$$

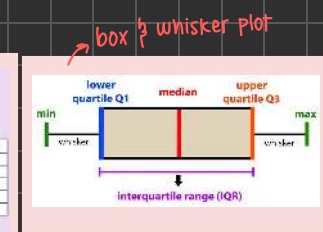
$$\bar{x}_{\text{coded}} = \frac{\sum(x - b)}{n} \quad \left\{ \begin{array}{l} \text{ungrouped} \\ \bar{x}_{\text{coded}}: \text{mean of coded} \\ a: \text{Scaling factor } (\times \text{ or } \div) \\ b: \text{Shift value } (+ \text{ or } -) \\ n: \text{total number of data points.} \end{array} \right.$$

$$* \bar{x} = a(\bar{x}_{\text{coded}}) + b$$

- #4: Cumulative frequency: Points are plotted at the class upper boundaries, helping in estimating medians, quartiles, & proportion of distribution
- #5: Box-and-whisker diagrams: Displays the minimum, lower quartile (Q1), median (Q2), upper quartile (Q3), and maximum of data Set.



cumulative frequency



box & whisker plot

Using Cumulative Frequency Graphs

Estimating Median

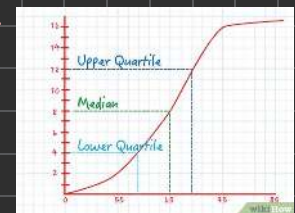
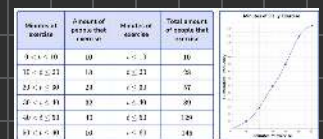
Find $\left(\frac{n}{2}\right)$ where n is total frequency on the y -axis. The value on the axis where $y = \left(\frac{n}{2}\right)$ is median.

Finding Quartiles

Lower Quartile (Q1): Find $\left(\frac{n}{4}\right)$ on y -axis and the corresponding x -axis value is Q1.

Upper Quartile (Q3): Find $\left(\frac{3n}{4}\right)$ on y -axis and the corresponding x -axis value is Q3.

The Interquartile Range: $Q3 - Q1$.



Measures of Variation

- #1: Range: maximum value in data Set minus the minimum Value.
- #2: Interquartile Range: $Q3 - Q1$
↳ spread of the middle 50% of data.
- #3: Standard deviation (σ): measures the spread of data around mean ($\bar{x}$). It quantifies how much data deviates from mean.

$$\text{Ungrouped: } \sigma = \sqrt{\frac{\sum(x - \bar{x})^2}{n}}$$

$$\sigma = \sqrt{\frac{\sum(x^2)}{n} - \left(\frac{\sum(x)}{n}\right)^2}$$

$$\text{Grouped data: } \sigma = \sqrt{\frac{\sum f x^2}{\sum f} - \left(\frac{\sum f x}{\sum f}\right)^2}$$

Estimating Percentiles

For k^{th} percentile, calculate $(kn/100)$, locate this on the y -axis, and project to x -axis.

Find Proportion Above/below or between Values

- for proportion below a value, locate value on x -axis, go to the corresponding y -value and divide it by n to find proportion.
- for proportion above a value, Subtract y value from n & divide by n (total frequency).
- between two values, find y -axis values, Subtract the Smaller value from larger value, divide the ans by n .

extra notes:

Probability & Statistics 5.2) Permutations And Combinations

Permutation

Arranging objects in a specific order

$$\text{Formula: } nP_r = \frac{n!}{(n-r)!}$$

n: total number of objects

r: number of objects to arrange

Key Feature: **Order Matters**

Combination

Selecting objects without considering the order

$$\text{Formula: } nC_r = \frac{n!}{(n-r)!r!}$$

n: total number of objects.

r: number of objects to select.

key feature: **Order doesn't Matter**

Factorials

a factorial, represented as $n!$, is product of all positive integers from n to 1

$$n! = n(n-1)(n-2) \dots (1)$$

$$0! = 1$$

the number of ways to arrange n items into n slots is $n! ({}^nP_n)$

Example:

3 books can be arranged in $3! = 6$ different ways.

Examples

#1: With Repetition

If some objects are identical, divide by the factorial of number of repetitions.

Example: total number of ways to arrange the word NEEDLESS

total letters: 8

Repeated letters: E = 3, S = 2

$$\text{Formula} = \frac{8!}{3! \times 2!} = 3360$$

#2: With Restrictions

• Case 1: Two specific people must stand next to each other.

- treat two people as single block
- If there are n objects, the total number of objects becomes (n-1).
- multiply (n-1)! with number of ways the two people can be arranged within the block (2!).

#3: Two specific people must NOT stand next to each other

- Calculate the total arrangements (n!) and subtract the cases where the two people are together.

Two Rows or groups

- Distribute object into group and arrange each group.
- Use combinations to select objects for each group, then multiply permutations for arrangement.

Example:

divide 6 people into two rows of three each

Step 1 - choose 3 for each first row: ${}^6C_3 = 20$

Step 2 - Arrange each row: ${}^3P_3 \times {}^3P_3 = 36$

$$\text{Total} = 20 \times 36 = 720$$

How to decide?

1. Does order matter?

- Yes: Permutation
- No: Combination

2. are all items being arranged or only a subset?

Subset: nP_r or nC_r

All items: Factorial

3. Are there multiple stages or restrictions?

- multiply for independent stages.
- Adjust for restrictions (treat them as grouped unit).

Probability & Statistics 5.3) Probability

Evaluating Probabilities

Probabilities of an event, E, occurring is expressed as:

$$P(E) = \frac{\text{Number of favourable outcomes}}{\text{Total number of outcomes}}$$

Key methods:

1. Enumeration of Equiprobable Events

- Count all equally likely outcomes
- Example: when dice is rolled, total number of outcomes is $6 \times 6 = 36$.

2. Using Permutation or combination

- Permutations: Use when order matters
- Combinations: Use when order doesn't matter
- Example: If 2 balls are drawn from a bag of 5, use ${}^5C_2 = 10$ to calculate probabilities.

Addition and multiplication of probabilities

Addition Rule (for mutually exclusive events A & B):

$$P(A \text{ or } B) = P(A) + P(B)$$

Multiplication Rule (for independent events A & B):

$$P(A \text{ and } B) = P(A) \times P(B)$$

The probability of A occurring given B has already occurred:

$$P(A|B) = \frac{P(A \text{ and } B)}{P(B)}, P(B) \neq 0$$

Sample spaces

Represent outcomes visually

Tree diagrams

Use to calculate probabilities for sequential events.

Exclusive and Independent Events

- Exclusive Events
- Events that cannot happen simultaneously
- Example: rolling a 2 & 5 on a single die.
 $P(A \text{ and } B) = 0$

- Independent Events
- The occurrence of one event doesn't affect the other.
- A and B are independent if:
 $P(A \text{ and } B) = P(A) \times P(B)$

Probability & Statistics 5.4) Discrete Random Variables.

Probability distribution table.

A discrete random variable (X) takes distinct values with specific probabilities. To create a Probability distribution table, follow these steps:

1. List the possible values that X can take
2. Assign probabilities to each of those values

Example: Dice is rolled

Possible values for X are 1, 2, 3, 4, 5, 6 and the probabilities are all $\frac{1}{6}$.

X	1	2	3	4	5	6
$P(X)$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$	$\frac{1}{6}$

Expectation $E(X)$ and Variance $Var(X)$

Expectation (Mean) $E(X)$: the expected value or average of the random variable, calculated as:

$$E(X) = \sum (x) \times P(x)$$

where X is the value and $P(x)$ is the probability

Variance $Var(X)$: A measure of how spread out the values of the random variable are. It is

calculated as: $Var(X) = E(X^2) - (E(X))^2$

where $E(X^2)$ is expected value of X^2 .

Binomial Distribution

The binomial distribution models situations where there are two possible outcomes (success or failure) over fixed number of trials n .

Notation $B(n, p)$ represents a binomial distribution where:

- n : Number of trials.
- p : Probability of success in trials.

Formula for Probability: the probability of getting exactly k successes in n trials is:

$$P(X=k) = \binom{n}{k} p^k (1-p)^{n-k}$$

Geometric Distribution

The geometric distribution models the number of trials until the first success in a sequence of independent trials, each with probability p of success.

- Notation: $Geo(p)$ represents the geometric distribution, where p : Probability of success.

Formula: $P(X=r) = p(1-p)^{r-1}$ for $r=1, 2, 3, \dots$
for having first success on the r^{th} trial

extra notes

Probability & Statistics 5.5) The Normal Distribution.

Normal distribution to model Continuous Variables

The normal distribution is used to model continuous random variables, meaning data that can take any value within a range (like height, weight or time). It is often represented by a bell shaped curve called a normal curve. This distribution is symmetrical (both sides look the same).

Using Normal Distribution table.

normal distribution table help us find the probability associated with a certain value of a normal random variable. These tables are used to look up the Z-score, which helps calculate probabilities.

Z-score is calculated by:

$$z = \frac{x - \mu}{\sigma}$$

Where X : value of variable
 μ : mean (average)
 σ : standard deviation

Solving Problems with normal distribution

You can solve problems involving the normal distribution, where X is a random variable that follows a normal distribution $N(\mu, \sigma^2)$. You might need to find the probability of a value falling above or below a certain point (eg: $P(X > x)$).

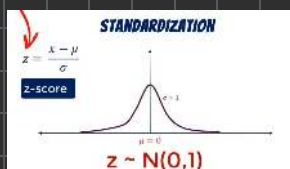
Example: Find the probability that a variable X is greater than 1 (eg: $P(X > 1)$) for a normal distribution with given mean and variance.

1. Standardize value to get Z-score
2. Use the Z-table to find the probability.

Standardization

Standardization is a process of converting a normal distribution into a standard normal distribution, which has a mean of 0 and standard deviation of 1. You can standardize using:

$$Z = \frac{x - \mu}{\sigma}$$



$$z = \frac{X - \bar{X}}{S}$$

where z is the standard score,
 S = the standard deviation of a sample,
 X = each value in the data set,
 $\bar{X}$ = mean of all values in the data set.

Normal distribution as an approximate to the binomial Distribution.

The normal distribution can approximate a binomial distribution when the number of trials n is large enough. This works when:

- $n \times p > 5$ where p is the probability of success
- $n \times q > 5$ and q is the probability of failure.

Continuity Correction: when approximating a binomial distribution using the normal distribution, a continuity correction is applied.

extra notes